

MARKABAT V4 — b!Bc Jc 'dF-bvDb' 'c

Repository: wagdee1115-blip/Wagggdi | Branch: markabat-v4-verification
bvDb!'c Jbã 2026-08-12

1. dV'c ' d 9dF* d 9dFJd¶'

bvDb!b!HbvEc 'dF*dç 4dö:e OdF*

npm install	' df,bÐ
npx prisma validate	' df,bÐ
npx prisma generate	' df,bÐ
git reset --hard wagggdi/markabat-v4-verification	' df,bÒ †%c-"bö) b!9d!Jdb %dFI 8864cbf)
git push wagggdi ... (commit bd26839)	' df,bÒ r Ö-prations + cascade + vitest.config
git push wagggdi ... (commit e231f9e)	' df,bÒ r %cVDbv- TS7023/7024 d J ledger.ts
GitHub Actions c-Dd' S#3 c-P	' df,bÒ r #ró#r .cvHb•
bV9bv/b' 66†VÖ ² -c A cascade migration + bÖ0d f—FW7Bæ6öæf-p	& p df,bÒ EbÖDd!Kbr r 6öÖÖ—B s"—c62 r DdR Jdö1d 9
npm run build (dV-dFJ)	' df,bÒ †Pxit 0) — 37 cVAbÖ)
npx tsc --noEmit (dV-dFJ)	' df,bÒ †Pxit 0 — cVAc #bæ7bv!)

dV' d 4d@

- b6HdB W6, AcFD b†3b†(token bæ'cv& (bÆDc6) c6'b†Bb•
- CI c-Dd' &C#cf3' AcFD b†3b†(TS7023/7024 d J ledger.ts
- CI c-Dd' ffcF6&b H341be07 d 4dF' b†3b†(c!Jbv(migration history

dV' d6'db \$Äô4'T@

dF' DB dV-dFJ d J b†Jbf) Replit — d6D bvDbv.b!(bv1bv* dfOd Pe 0bç -cV1d!Kl

2. dV'c ' d 9dF* d 9dFJd¶'

backend/lib/ledger.ts	bV6bvAb' ÆVFvW\$VçG y + DoubleEntryResult + e Type-only return type — bÖD TS7023/7024	' pushed (e231f9e)
backend/lib/otp.ts	b!Bc6JdR G ansaction dF-d 8 c-/bv/ bvDdV-bvHdF' Business Logic	' pushed (bÆDc6) c6'b†Bb•
backend/lib/transfer-workflow.ts	bV2bvDb' 76W tPayoutAccount bvDdV(d61b' Bb†C Business Logic	' pushed (bÆDc6) c6'b†Bb•
backend/lib/government.ts	bV6bvAb' &WGW n type cV1d!- dF@ getElectronicI Type-only	' pushed (bÆDc6) c6'b†Bb•
backend/lib/identity-verification.ts	b!Fd Jc ,dVJc' ÖWF†öG2 Dd • &ðvider interface Type-only	' pushed (bÆDc6) c6'b†Bb•
backend/lib/malware-scanner.ts	bV5dF'bÖ Fd†9 clean property Type-only	' pushed (bÆDc6) c6'b†Bb•
backend/lib/payment-provider.ts	bV5dF'bÖ &WGW n type dFJcFEdB 7F GW3 Payn Type-only	' pushed (bÆDc6) c6'b†Bb•
backend/lib/providers/phone.ts	bV5dF'bÖ ÖWF†öB 6-væ GW&W2 Db!7bv(d" †öæ Type-only	' pushed (bÆDc6) c6'b†Bb•
backend/lib/providers/registry.ts	bV5dF'bÖ Öö0kPhoneProvider import Type-only	' pushed (bÆDc6) c6'b†Bb•
backend/app/api/auctions/[id]/bids/route.ts	bV6bvAb' àull guard d&(dB 'dFHcVHdB Dbæ5bv&c! Bug fix	' pushed (bÆDc6) c6'b†Bb•

backend/prisma/migrations/20260812010041_init/migration.sql	Initial migration (56K)	Migration	' pushed (bd26839)
backend/prisma/migrations/migration_lock.toml	Prisma migration lock file	Config	' pushed (bd26839)
backend/prisma/schema.prisma + onDelete: Cascade (4 models)	b6OcfJd Ad¢ &C#cf3' +dR #dö9d!/ dV-dFJd¶' d J º Migration/Schema		& p local only — remote d!-b!Hd¢ Cascade
backend/prisma/migrations/20260812040000_cascade_sale_children/	b6OcfJd Ad¢ &C#cf3' +dR -dö0d EbÖDd!Kbr Ad¢ Migration		& p local only — remote d!-b!Hd¢ Gc G migration
backend/vitest.config.ts	b6OcfJd ‡ ööÃ forks) d J bd26839 b¶E bÖOc A dV Config		& p local only — remote d!-b!Hd!G

3. Git

bvDb6HbvEc Hbæ1bÆG

```
git branch --show-current
! '
markabat-v4-verification
git status --short
! ' (dF'
output - working
tree df8d!A)
git rev-parse HEAD
! ' a79
9f3c9c96dbfc004c7b4
5e1b8da5ff9530953f

git log -10 --oneline:
a799f3c Revert migration and remove test configuration
! • HEAD (dV-dFJ)
1db49a3 Initialize project configuration and build cache
! • dV-dFJ e231f9e
fix(ts): add explicit return type ...
TS7023/7024
! • wagggdi/markabat-v4-verification
bd26839
fix(ci): add Prisma migrations, cascade deletes ...
8864cbf
fix(ci): remove unresolved npm cache path ...
341be07
fix(v4): verify with Prisma migrations in CI
ba68eaf Align duplicate payment test with idempotent ...
8f16835 Persist OTP failed-attempt counters before errors
5b7f5cf Fix concurrent double-entry idempotency locking
15f84d5 Make V4
```

```
external E2E
provider gate
explicit

git remote -v:
  gitsafe-backup
git://gitsafe:5418/
backup.git
  origin
https://github.com/
wagdee1115-blip/
Extract-MARKABAT-
V4 !• bæ'cv&
  wagggdi h
ttps://...@github.c
om/wagdee1115-blip/
Wagggdi.git !• cV-d!-
dVDhæ5 hÖ'dF) Git
```

Local HEAD SHA	a799f3c
dV-b!Hd' Æö6 Å „T @	Revert: bÖ0d 6 66 FR Ö-pration + bV9bv/b' 66†VÖ ç isma + bÖ0d f—FW7Bæ6öæf-p
dvD pushað	'L dF' — dV-dFJ d Bcp
b&.c 4,, Ec Ad†9	e231f9e (wagdee1115-blip/Wagggdi)
bvDd 1c' 'dFEc Ad†9 bVDd!G	markabat-v4-verification
bvDd &VÖ÷FR 'dF5bÖJbÐ	wagggdi !' wagdee1115-blip/Wagggdi.git
origin (bæ'cv&)	Extract-MARKABAT-V4 !• dF' b!3b!.böEdp
Local b6EbV remote b†@	2 commits (1db49a3, a799f3c)
PR dVAb!HbÐ	'L dF' d!HbÆ/

4. Prisma

prisma validate	' PASS	' PASS (CI)
prisma generate	' PASS	' PASS (CI)
Initial migration	' 20260812010041_init/migration.sql	' dVHbÆHbð
bÖ,dR Ö-pration.sql	56,493 bytes (56K)	56,493 bytes
migration_lock.toml	' dVHbÆHbð	' dVHbÆHbð
migrate deploy	'L dFE d!OcF:dæQdB †Dbr D" EbÖDd☛	' PASS (CI)
migrate status	'L dFE d!OcF:dæQdB †Dbr D" EbÖDd☛	' PASS (CI)
schema.prisma — onDelete	0 dV1b' †#dö9d!)	4 dV1bv* Cascade (dV' c&dB Ad☛ &VÖ÷FR•
cascade migration 20260812040000	'L dV-c Hd EbÖDd!Kbp	' dVHbÆHbð Ad☛ &VÖ÷FP

5. PostgreSQL

DB dV-dFJb' Ad☛ &W Æ—@	'L dF' b!*d†Ac r DdR *döAc—Ne D
bv*cV'dB EbÖDd	'L dFE d!ObÆ1dæQb€
migrate deploy d J CI	' PASS c—Dd' S#3 c—R ... ostgreSQL container)
migrate status d J CI	' PASS c—Dd' S#3 c—P
Schema up to date d J CI	' (c—Dd' S#3 c—R r JcFEdB 6 66 FR Ö-pration)

6. TypeScript

```
$ cd backend && npx tsc --noEmit
(dF' output)
TSC_EXIT: 0
' PASS — cVAc #bæ7bv!.bvDd6Hbð 'dFEbÖDd☛ ,,,T B s"—c62' JdV1 b†/d†F b6J bæ?
```

7. Tests

bvDdV5bö1: GitHub Actions c–Dd' 6öÖÖ—B S#3 c–R r Dbr D" EbÖDd¢ Ad¢ &W Æ—F				
Full Suite	8 dVDd 'b	' PASS	24 passed (24)	Cl e231f9e
OTP	tests/otp.test.ts	' PASS	1	—
Financial	tests/financial.test.ts	' PASS	2	—
Sale Concurrency	tests/sale-concurrency.test.ts	' PASS	1	—
Security	tests/security.test.ts	' PASS	6	—
State Machine	tests/state-machine.test.ts	' PASS	2	—
Core	tests/core.test.ts	' PASS	6	—
E2E	tests/e2e.test.ts	' PASS	2	—
Auction	tests/auction.test.ts	' PASS	4 (cfEdb gV/ÆÂ 7V—Fl	—
Authorization	c Jc Ed†,d†/ d6EdFA dV3b Bd@	N/A	—	dF' d Hb/Æ/ authorization.test.ts
Direct Sale	c Jc Ed†,d†/ d6EdFA dV3b Bd@	N/A	—	dF' d Hb/Æ/ direct-sale.test.ts
Auto-Bid	c Jc Ed†,d†/ d6EdFA dV3b Bd@	N/A	—	dV:cv cfEdb V7F—ôâçFW7BçG0

8. Build / Runtime

```
$ cd backend && npm run build
> markabat@1.2.0 build
> next build

' Compiled successfully
' Linting and checking validity of types
' Collecting page data
' Generating static pages (37/37)
' Finalizing page optimization

BUILD EXIT: 0
```

npm run build	' PASS — exit 0 — 37 cVAbÖ)
npm start	'L dFE d OcF:daeQdB r Dbr porkflow dVOd6HdaeQdf dF' DB dV-dFJ
bvDb 7b†Jd" (bö# d 9dFJd¶ 'aö	'L dF'
Runtime b6.cv'b	c Jc Ec–1d†Ab' r 'dF*cv(d B dFE d (bö#

9. External Providers

SMS	'L NOT_CONFIGURED	SMS_PROVIDER_URL, SMS_PROVIDER_SECRET
Payment	'L NOT_CONFIGURED	PAYMENT_PROVIDER_URL, PAYMENT_PROVIDER_WEBHOOK_SECRET
Escrow	'L NOT_CONFIGURED	ESCROW_PROVIDER_URL, ESCROW_PROVIDER_SECRET
Payout	'L NOT_CONFIGURED	PAYOUT_PROVIDER_URL, PAYOUT_PROVIDER_SECRET
Government/Traffic	'L NOT_CONFIGURED	TRAFFIC_PROVIDER_URL, TRAFFIC_PROVIDER_SECRET
Identity/KYC	'L NOT_CONFIGURED	IDENTITY_PROVIDER_URL
Storage	'L NOT_CONFIGURED	STORAGE_PROVIDER_URL, STORAGE_PROVIDER_TOKEN
Malware Scanner	'L NOT_CONFIGURED	MALWARE_SCANNER_URL, MALWARE_SCANNER_SECRET

AI Description	'L NOT_CONFIGURED	AI_DESCRIPTION_PROVIDER_URL
Plate Redaction	'L NOT_CONFIGURED	PLATE_REDACTION_PROVIDER_URL
Bank Transfer	'L NOT_CONFIGURED	BANK_PROVIDER_URL, BANK_TRANSFER_ACCOUNT_ID
Auction Deposit	'L NOT_CONFIGURED	AUCTION_DEPOSIT_PROVIDER_URL
Authorization Doc Storage	'L NOT_CONFIGURED	AUTHORIZATION_DOCUMENT_STORAGE_URL

b/ÆEd!9 bvDd &ðviders b!OdFBd¢ ääT_CONFIGURED:*_PROVIDER_REQUIRED c-
silent fallback.

10. GitHub Actions

Workflow dVHb/ÆHbö	' df9dR r %`erify MARKABAT V4"
b&.c un	e231f9e — 2026-08-12T05:32:28Z
bvDdf*d!,b•	' PASS — 27/27 bæ7d†)
b6HdB .cvHb' AcFDb	dF' d!Hb/Æ/ (d6D bvDbæ7d†'b¢ Fb/Æ-b¢•
dvD c6Jdö4c!Ne D dV,bö/d¶!aö	'L dF' — bvDd " 6öÖÖ—G2 'dFEbÖDd!) dFE b!Oc Ac•

e231f9e	' success	d6D bvDbæ7d†'b¢ Fb/Æ-b¢ r #ró#p
bd26839	'L failure	TS7023/7024 d J ledger.ts
8864cbf	'L failure	c!Jbv(migration history
341be07	'L failure	c!Jbv(migration history

11. Vercel!

bV9bö'bò Vercel	'L NOT READY — dF' d!Hb/Æ/ b!Cd†Jd`
Build verifiedaö	' df9dR †ç Ò un build d!Fb/Æ- dV-dFJd¶!)
DATABASE_URL d J Production	'L c!Jc Edö9d!Ne F
JWT_SECRET d J Production	'L c!Jc Edö9d!Ne F
b/ÆEd!9 bvDd &ðviders	'L c!Jc EdöCd†Ne Fb' f 2 .böEb•
Environment Variables b/Æ'dv2b—	'L dF' — d6Ddv' d 'c :b•

12. bvDb/Æ'dv2 bvDb&F (dV+b†* d 9dFJd¶!)

- ' TypeScript — cVAc #bæ7bv! (tsc --noEmit exit 0)
- ' Production Build — next build d!Fb/Æ- (37 cVAböÖ))
- ' Prisma schema — cV-d!- d†Edö*bÖBd" EdfG
- ' Initial migration 20260812010041_init — dVHb/ÆHbö HdVOcv(dæQd" Ad¢ 4•
- ' GitHub Actions workflow — dVHb/ÆHbö HdVOD6HdæQd`
- ' 24 bv.b!(bv1 — d6Ddv' df,bÖ* d J Cl c—Dd' 6öÖÖ—B S#3 c—P
- ' OTP logic — c—bv/ bvDdV-bvHdF'b¢ Jdö-d 8 böÖ*d' 9df/ bvDd 4d@
- ' Ledger double-entry — advisory lock d!9dVD bö'bæD explicit transaction
- ' TypeScript interfaces — 9 dVDd 'b¢ EcVDbÖ)

13. bvDdf'd&5 bvDb&F

- 'L 2 commits dV-dFJb' :d!1 dV1d Hc—) (a799f3c, 1db49a3) b!*c—'c 6 dV9 remote
- 'L dF' PR dVAb!HbÖ r Dbr Ec6'c DdF@ review b6H bvDböEbÀ
- 'L origin d!4d!1 bVDd' 'dFEc6*d†/c' 'dF.bv7bb „W‡G act-MARKABAT-V4)
- 'L b/ÆEd!9 bvDd 2 Pexternal provider c!Jc EdöCd†Ne Fb•
- 'L dF' DB d J b†Jbf) Replit — bvDbv.b!(bv1bv* dF' b!OcF:dæQdB EbÖDd!Kbp
- 'L dF' Vercel deployment — dF' b!Cd†Jd`
- 'L DATABASE_URL d, ¥uEö4T5\$UB H CRON_SECRET d J production — c!Jc Edö9d!Ne
- 'L Payout account guard — bvDb†'bf9 d!Ed6Fdr %df4bv! b†Jc' (böHdb -c6'b, yout

14. bvDdV.bv7c

& p 2 commits dV-dfJb' :d!1 dV1d Hc-)	a799f3c + 1db49a3 — bvDd /Æö6 Â #dV'dR &VÖ÷FR (d " 6öÖÖ—G0
& p b!9bv1cb 66†VÖ (d!F local d‡&VÖ÷FP	Local: dF' Cascade. Remote e231f9e: Cascade c—Dd' B ÖöFVÇ0. c Ac' s"—c62 Bbò Jd63c 4•
& p vitest.config.ts dV-c Hd EbÖDd!Kbp	Remote d!-b!Hd!G — bÖ0d G d!Od Bbò ööÃ forks d†Bbò *b! /bv.dB 'dF'bæ*b†'c 'b
& p origin remote bæ'cv&	d!4d!1 bVDd' W‡G act-MARKABAT-V4 — b6J push b†/d†F b!-böJbò &VÖ÷FR 3d!0dv(dFEc6*d†/c' .bv7b`
& p Token d J URL bvDd &VÖ÷FP	wagggdi remote d!-b!Hd¢ Fö°en d J URL dV1bfJ d J git remote -v
'L dF' PR	dF' dV3bv1 c 3dVJ dFDd &Pview b6H bvDböEbÂ Ad¢ Ö —à
'L b/ÆEd!9 providers c!Jc EdöCd†Ne Fb•	b6J c—EdFJb' (d!9 b6H dV5bv/d&) b6H b!-d†JdB EdFCd!) c6*d 4dB Ad¢ &öGV7F—öà
'L Cascade migration d J remote — schema	c—/dR *cv'b†B b†Jdb Ö—pration bvDdV1d Hc-) d†'dF@ schema bvDdV-dfJ

15. bvDbÖCdR 'dFFdv'bfJ

PARTIALLY VERIFIED

- CI df,bÖ CbvEdFKbr f#rö#r .cvHb" 9dFI commit e231f9e d J wagdee1115-blip/Wagggdi — TypeScript, migrations, b/ÆEd!9 bvDbv.b!(bv1bv* b
- dFCdb 'dF-bvDb' 'dFEbÖDd!) b!.bvDd 'dF@ remote: bvDd " 6öÖÖ—G2 'dFEbÖDd!) (a799f3c + 1db49a3) dFE b!Oc Ac– d†*bÖ0d f—FW7B
- dF' PR dVAb!HbÒ r Dbr Ec6'c 1c6Ed¢ DböEbÂ 'dFCd†/ bvDdV*bÖBd" EdfG d J main.
- b/ÆEd!9 bvDd Pxternal providers (13 dVFdv') c!Jc EdöCd†Ne Fb' r 'dF*cv(d!B dF' d!Ed6Fdr %b!EbvE b6J c—EdFJb' (d!9 b6H dV5bv/d&) b6H
- bvDb!9bv1cb (d!F local d‡&VÖ÷FR JdV+dB Ebæ'cv1b" c Ac' 6öÖÖ—G2 s"—c62 3d!Cc61 bvDbÖ'dF) bvDb!J b/Æ9dF* CI d!Fb/Æ- (FK violation

